Breast Ultrasound Image Classification
Using Deep Learning

Handling Multi-Class Imbalance on the BUSI Dataset

Submitted by

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Abstract

Breast cancer is one of the leading causes of death among women worldwide. Early and accurate detection significantly improves patient survival rates. This project develops a deep learning-based system to classify breast ultrasound images into three categories — *benign*, *malignant*, and *normal* — using the publicly available BUSI dataset. A pretrained ResNet50 model is employed as the backbone through transfer learning. Six strategies are systematically explored to handle the inherent class imbalance: Baseline, Data Augmentation, Weighted Random Sampling, Focal Loss, SMOTE Hybrid, and Mixup Training. All models are trained in PyTorch and evaluated on a held-out test set of 117 images using accuracy, precision, recall, and F1-score.

This report presents the methodology, experiments, results, and analysis of the BUSI breast ultrasound image classification project.

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All experiments use ResNet50 pretrained on ImageNet.
Dataset: BUSI — 780 breast ultrasound images across 3 classes.

1. Introduction

Breast cancer is one of the most common and life-threatening cancers among women worldwide. Early and accurate detection is critical for improving patient outcomes and survival rates. Ultrasound imaging is widely preferred in clinical settings because it is non-invasive, radiation-free, inexpensive, and widely accessible. However, manual interpretation requires significant radiological expertise and is subject to inter-observer variability.

The rise of deep learning, particularly Convolutional Neural Networks (CNNs), has opened a promising avenue for automated and consistent medical image diagnosis. In this project, a CNN-based classifier is built using transfer learning with ResNet50. A central challenge is the significant class imbalance in the BUSI dataset, where benign cases far outnumber malignant and normal ones. Six different imbalance-handling strategies are explored and compared.

2. Problem Statement

The BUSI dataset is inherently imbalanced: benign cases account for over 55% of images while normal cases make up only 17%. A naive classifier tends to be biased toward the majority class, achieving high accuracy while performing poorly on minority classes. In a clinical context, failing to detect a malignant case (false negative) has severe consequences.

Project Goals:

1. Develop an accurate deep learning model to classify breast ultrasound images into benign, malignant, and normal.
2. Systematically compare six imbalance-handling strategies to determine which best improves minority class sensitivity.

3. Dataset Description

The **BUSI (Breast Ultrasound Images)** dataset is a publicly available benchmark collected from 600 female patients at Badr University Hospital in 2018. It contains 780 labelled ultrasound images across three classes. Segmentation mask files are excluded from this project.

Class	Total	Proportion	Train (70%)	Test (15%)
Benign	437	**55.8%**	305	66
Malignant	210	26.9%	147	31
Normal	133	**17.0%**	93	20
Total	**780**	**100%**	**546**	**117**

Table 1: BUSI dataset class distribution. **Green** = majority class (benign, 55.8%); **Amber** = minority (malignant, 26.9%); **Red** = severe minority (normal, 17.0%).

Class Imbalance Warning: The benign class (437 images) is **3.3**× larger than normal (133 images) and **2.1**× larger than malignant (210 images). A naive model will be strongly biased toward predicting "benign", achieving high accuracy while failing on the clinically critical minority classes.

A **stratified 70/15/15** split preserves class proportions across training, validation, and test sets. All images are converted to RGB, resized to 224×224, and normalised using ImageNet statistics.

4. Methodology

The complete pipeline from raw data to final comparison is described below.

4.1. Data Preparation

Image paths are collected by walking the three class folders (`benign/`, `malignant/`, `normal/`). Mask files (filenames containing `_mask`) are explicitly excluded. Integer labels are assigned: be-

nign=0, malignant=1, normal=2. A `pandas` DataFrame stores all path-label pairs.

A **stratified split** using `train_test_split` (with `random_state=42`) produces:

- **Training set (70%):** 546 images used for model optimisation
- **Validation set (15%):** 117 images used for early stopping and checkpoint selection
- **Test set (15%):** 117 images held out entirely until final evaluation

4.2. Data Loading and Transforms

A custom PyTorch `Dataset` class (`UltrasoundDS`) handles image loading. Each image is opened with PIL, converted to RGB (ensuring grayscale images become 3-channel), and converted to a NumPy array for Albumentations. Two transform pipelines are defined:

- **Plain transform:** Resize to 224×224 + ImageNet normalisation only. Used for validation, test, and baseline training.
- **Augmented transform:** Adds horizontal flip (p=0.5), rotation $\pm 20°$ (p=0.5), random brightness/contrast (p=0.4), and Gaussian blur (p=0.3) before normalisation. Used for augmentation, weighted sampling, focal loss, and mixup experiments.

All transforms output a normalised PyTorch tensor of shape $[3, 224, 224]$.

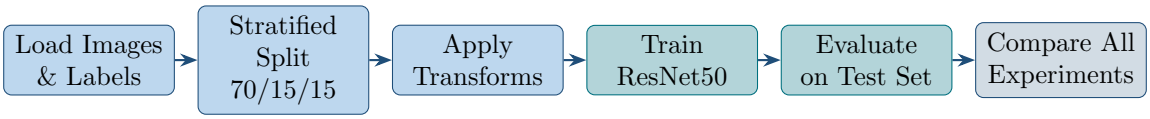
4.3. Training Procedure

Each experiment follows the same training loop structure:

1. **Forward pass:** Batch of images passed through ResNet50 to produce logits.
2. **Loss computation:** Loss function applied to logits and ground-truth labels.
3. **Backward pass:** Gradients computed via backpropagation.
4. **Parameter update:** AdamW optimiser updates model weights.
5. **Validation:** After every epoch, model evaluated on validation set.
6. **Checkpointing:** Best model weights (lowest validation loss) saved and restored at end of training.

Training runs for **8 epochs** with **batch size 16** and **AdamW** (lr $= 10^{-4}$, weight decay $= 10^{-4}$). GPU (CUDA) is used when available.

4.4. Pipeline Overview



5. Model Architecture

5.1. ResNet50 Overview

ResNet50 is a 50-layer deep residual network introduced by He et al. (2016). The key innovation is the *residual connection* (skip connection), which allows gradients to flow directly through the network by adding the input of a block to its output:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}$$

This mitigates the vanishing gradient problem that plagues very deep networks, enabling effective training of 50+ layer architectures.

ResNet50 is organised into five stages:

Stage	Component	Output Size	Params
Input	$224 \times 224 \times 3$ image	$224 \times 224 \times 3$	—
Stage 1	Conv 7×7, stride 2 + MaxPool	$56 \times 56 \times 64$	9K
Stage 2	$3 \times$ Bottleneck blocks	$56 \times 56 \times 256$	212K
Stage 3	$4 \times$ Bottleneck blocks	$28 \times 28 \times 512$	1.2M
Stage 4	$6 \times$ Bottleneck blocks	$14 \times 14 \times 1024$	7.1M
Stage 5	$3 \times$ Bottleneck blocks	$7 \times 7 \times 2048$	14.9M
Pool	Global Average Pooling	$1 \times 1 \times 2048$	—
Head	**Linear**$(2048, 3)$	3	6K

Table 2: ResNet50 layer structure with output sizes. The original FC(1000) head is replaced with FC(3).

5.2. Transfer Learning Strategy

The model is initialised with **ImageNet pretrained weights**. The entire network (including all convolutional stages) is fine-tuned end-to-end, not frozen. Only the final classification head is replaced:

$$\underbrace{\text{ResNet50 Backbone}}_{\text{pretrained, fine-tuned}} \longrightarrow \text{Global Avg Pool} \longrightarrow \underbrace{\text{Linear}(2048,\ 3)}_{\text{new head}} \longrightarrow \text{Softmax}$$

Fine-tuning all layers allows the convolutional features to adapt from natural images (ImageNet) to medical ultrasound textures, which differ significantly in visual characteristics. The pretrained weights serve as a strong initialisation that accelerates convergence and improves generalisation on the relatively small BUSI dataset.

5.3. Training Configuration

Optimiser	AdamW (adaptive learning rate with weight decay)
Learning Rate	1×10^{-4}
Weight Decay	1×10^{-4}
Epochs	8 (best val-loss checkpoint restored)
Batch Size	16
Hardware	GPU (CUDA) when available, else CPU

6. Handling Class Imbalance Techniques

6.1. Baseline

Standard CrossEntropy loss with random shuffling and no augmentation. Serves as the reference point showing the raw effect of class imbalance on model performance.

6.2. Data Augmentation

Medical-style augmentations applied via Albumentations at training time: horizontal flip (p=0.5), rotation $\pm 20°$ (p=0.5), brightness/contrast (p=0.4), and Gaussian blur (p=0.3). Increases training diversity without collecting new data.

6.3. Weighted Random Sampling

Each training sample is assigned weight $w_c = N/N_c$ (inverse class frequency). PyTorch's `WeightedRandomSampler` ensures each mini-batch contains a balanced class representation, effectively over-representing the minority classes at the sampling level.

6.4. Focal Loss

Modifies cross-entropy to down-weight easy majority examples and focus on hard minority samples:

$$FL(p_t) = -(1 - p_t)^{\gamma} \log(p_t), \qquad \gamma = 2$$

When the model confidently classifies a sample ($p_t \to 1$), the term $(1 - p_t)^\gamma$ approaches zero, reducing its loss contribution. Hard or misclassified minority samples retain high loss weight.

6.5. SMOTE Hybrid

ResNet50 (with head removed) acts as a frozen feature extractor producing 2048-d embeddings for all training images. SMOTE synthesises new minority class feature vectors by interpolating between existing K-nearest neighbours in this feature space. A lightweight MLP (512→256) is trained on the balanced feature set, decoupling representation learning from classification.

6.6. Mixup Training

Pairs of training images and labels are interpolated using $\lambda \sim \text{Beta}(0.4, 0.4)$:

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j, \qquad \tilde{y} = \lambda y_i + (1 - \lambda) y_j$$

Training on soft-labelled mixed images encourages smoother decision boundaries and reduces over-confident majority class predictions.

7. Evaluation Metrics

- **Accuracy:** Fraction of correct predictions. Misleading on imbalanced datasets.
- **Precision (Macro):** Average per-class fraction of positive predictions that are correct.
- **Recall (Macro):** Average per-class fraction of actual positives correctly identified. *Most clinically critical* — high malignant recall means fewer missed cancers.
- **F1-Score (Macro):** Harmonic mean of precision and recall. Primary comparison metric. All macro averages weight each class equally regardless of support.

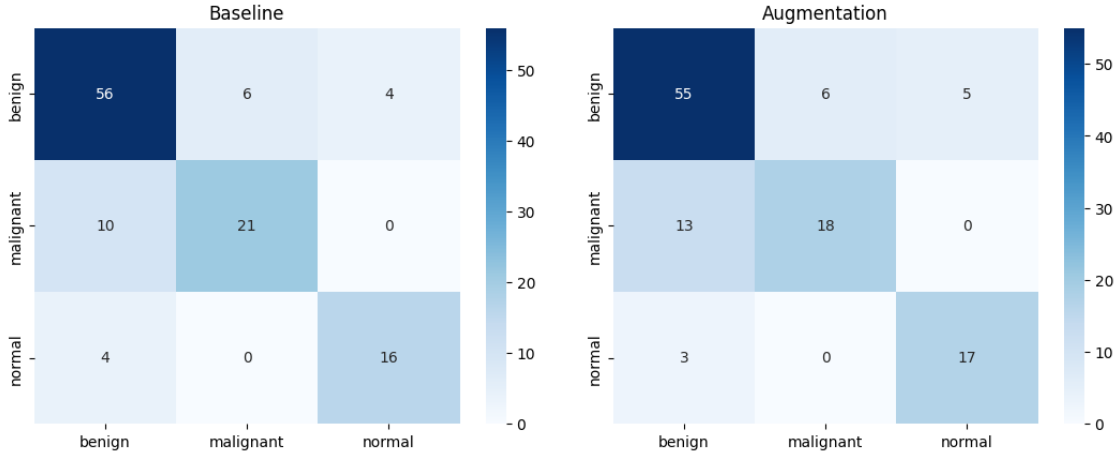
8. Results and Analysis

8.1. Per-Class Classification Results

Experiment	Class	Precision	Recall	F1
Baseline	Benign	0.80	0.85	0.82
	Malignant	0.78	0.68	0.72
	Normal	0.80	0.80	0.80
Augmentation	Benign	0.77	0.83	0.80
	Malignant	0.75	0.58	0.65
	Normal	0.77	0.85	0.81
Weighted Sampling	Benign	0.82	0.80	0.81
	Malignant	0.69	0.77	0.73
	Normal	0.82	0.70	0.76
Focal Loss	Benign	0.82	0.77	0.80
	Malignant	0.73	0.77	0.75
	Normal	0.73	0.80	0.76
SMOTE Hybrid	Benign	**0.85**	0.80	**0.83**
	Malignant	0.67	0.77	0.72
	Normal	0.74	0.70	0.72
Mixup	Benign	0.79	0.73	0.76
	Malignant	0.63	0.77	0.70
	Normal	0.78	0.70	0.74

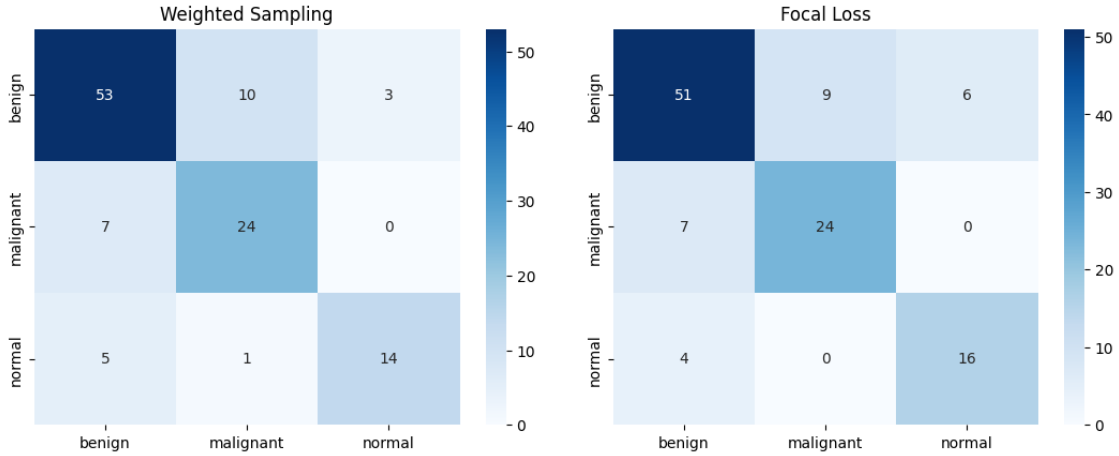
Table 3: Per-class results on the test set. Green ≥ 0.80 Amber 0.70–0.79 Red < 0.70

8.2. Confusion Matrices



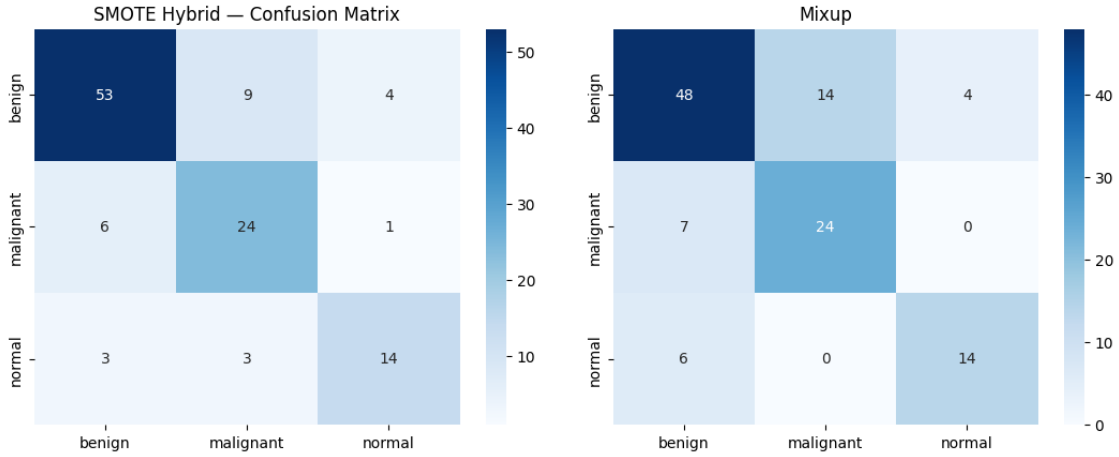
(a) Baseline

(b) Augmentation



(c) Weighted Sampling

(d) Focal Loss



(e) SMOTE Hybrid

(f) Mixup

Figure 1: Confusion matrices for all six experiments on the test set.

8.3. Overall Comparison

Experiment	Accuracy	Precision	Recall	F1 Score
Baseline	**0.7949**	**0.7926**	0.7753	**0.7826**
Augmentation	0.7692	0.7658	0.7547	0.7557
Weighted Sampling	0.7778	0.7749	0.7591	0.7644
Focal Loss	0.7778	0.7590	**0.7823**	0.7696
SMOTE Hybrid	0.7778	0.7528	0.7591	0.7542
Mixup	0.7350	0.7321	0.7338	0.7295

Table 4: Macro-averaged metrics on the test set. Green ≥ 0.78 Amber 0.74–0.77 Red < 0.74. Bold = best per column.

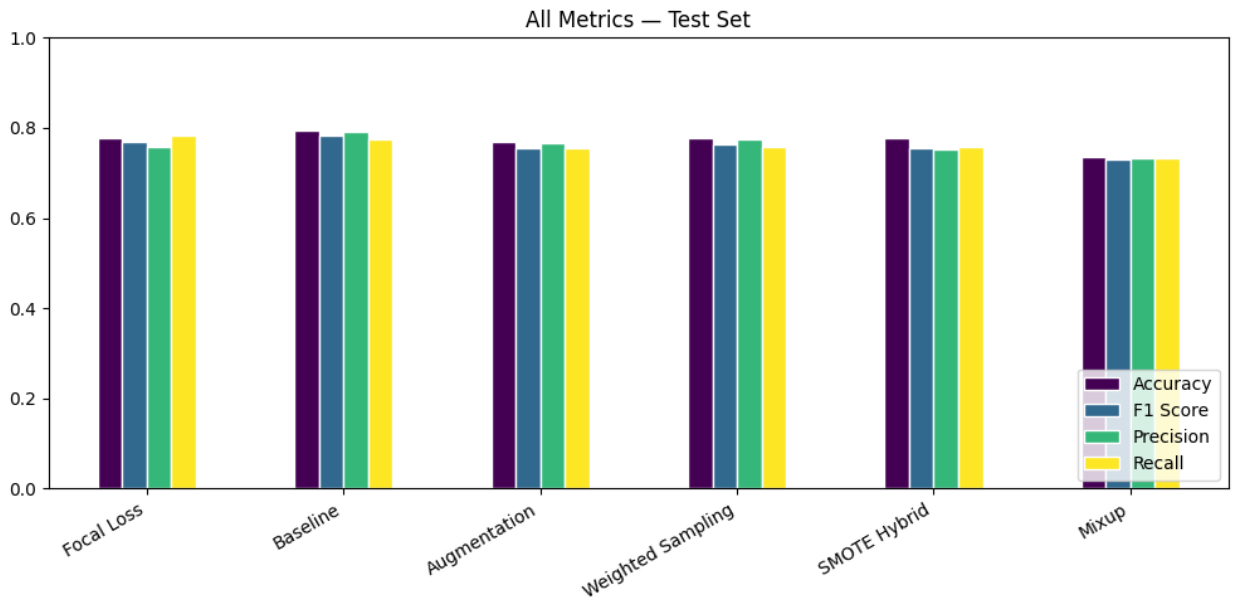


Figure 2: Comparison of all evaluation metrics across the six experiments.

9. Discussion

Why accuracy is misleading. A model always predicting "benign" achieves >56% accuracy while being clinically useless. Macro F1 and per-class recall are far more informative for imbalanced datasets.

Baseline performs strongly overall. The pretrained ResNet50 achieves the highest accuracy (0.7949) and F1 (0.7826), confirming that ImageNet features transfer well to ultrasound images. However, its malignant recall is the lowest (0.68) — it misses 32% of actual malignant cases, which is unacceptable in clinical screening.

Focal Loss best addresses clinical recall. With macro recall of 0.7823 and malignant recall of 0.77, Focal Loss achieves the most important improvement by directing gradient toward hard minority samples.

Weighted Sampling is equally effective for malignant recall (0.77), requiring only a dataloader-level change — making it the simplest effective technique.

SMOTE Hybrid achieves the best benign precision (0.85) in feature space but does not surpass end-to-end models on overall metrics, as feature-space classification loses fine pixel-level discriminability.

Mixup underperforms (F1: 0.7295). For ultrasound images where discriminating features are subtle textures, blending unrelated images creates ambiguous training samples that hurt generalisation.

10. Conclusion

This project systematically studied six class imbalance handling strategies for breast ultrasound classification using ResNet50 on the BUSI dataset. Key findings:

- **Baseline** achieved the best overall accuracy (0.7949) and F1 (0.7826).
- **Focal Loss** achieved the highest macro recall (0.7823) and improved malignant recall from 0.68 to 0.77 — the most clinically meaningful result.
- **Weighted Sampling** equally improved malignant recall to 0.77 with minimal implementation overhead.
- **Mixup** was the least effective technique for this dataset.
- **Accuracy alone is insufficient** — macro F1 and per-class recall must be reported for imbalanced medical datasets.

Recommendation: For clinical deployment, **Focal Loss combined with Weighted Sampling** is recommended as it best prioritises sensitivity on minority (malignant and normal) classes. Future work could explore ensemble methods, self-supervised medical pretraining, and class-specific threshold calibration.